ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT

Department of Electronics and Computer Science (ECS)

PLC AND HMI COURSE

DAY 4,5,6 REPORT

Industrial OEE Monitoring and Control System

Mitsubishi FX5U PLC (GX Works3) | GOT HMI (GT Designer3 / GT Simulator3)

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Table of Contents

1. Introduction 4
1.1 Purpose of this report 4
1.2 Aim 4
1.3 Objectives 4
1.4 Scope and limitations 4
2. Background Theory 5
2.1 Programmable Logic Controller (PLC) 5
2.2 Mitsubishi FX5U and GX Works3 5
2.3 Ladder logic 5
2.4 Instructions used in the program 5
2.5 Human-Machine Interface (HMI) 6
2.6 Types of HMI objects used 6
3. Overall Equipment Effectiveness (OEE) 7
3.1 What is OEE? 7
3.2 The three factors 7
3.3 Scaling used in the PLC (per-mille) 7
3.4 Illustrative worked example 7
4. System Description 9
4.1 Hardware and software 9
4.2 System architecture 9
4.3 Operating sequence 9
4.4 Operating states 10
5. PLC Device Allocation 12
5.1 Physical inputs (reserved for later) 12
5.2 Outputs 12
5.3 HMI command bits 12
5.4 Alarm and fault bits 13
5.5 Status bits 13
5.6 Internal pulses and special relays 13
5.7 Data registers 13
5.8 Devices marked not used 14
5.9 HMI screens 14
5.10 Test inputs versus real inputs 15
6. Ladder Program Description 16
6.1 Initialisation, mode selection and faults (page 1) 17
6.2 System reset, outputs and counting (page 2) 19
6.3 Target and OEE calculation (pages 3 and 4) 21
6.4 Summary of all rungs 22
7. OEE Calculation in the PLC 24

7.1 Overview 24
7.2 Availability (steps 273 to 304) 24
7.3 Performance (steps 304 to 356) 24
7.4 Quality (steps 356 to 386) 25
7.5 Overall OEE (steps 386 to 440) 25
7.6 Protection against zero values 25
7.7 Register summary 26
8. HMI Design 27
8.1 Design approach 27
8.2 Screen list 27
8.3 Screen 1: Home 27
8.4 Screen 2: Production Monitor 28
8.5 Machine Status page (test buttons) 29
8.6 Screen 3: Settings / Modes 30
8.7 Screen 4: Alarm / Fault 31
8.8 Navigation 31
8.9 Documentation consistency 31
9. PLC and HMI Integration 32
9.1 Data exchange 32
9.2 Why momentary buttons are used 32
9.3 Example sequence 32
10. Testing 33
10.1 Test setup 33
10.2 Status recorded in the project notes 33
10.3 Test plan for the final program 33
11. Observations from Ladder and HMI Review 35
12. Safety and Limitations 36
13. Learning Outcomes 37
14. Conclusion and Future Scope 38
14.1 Conclusion 38
14.2 Future scope 38
15. References 39

1. Introduction

1.1 Purpose of this report

This report documents the work carried out on Day 5 of the PLC and HMI course. The work on this day was the development of an **Industrial OEE (Overall Equipment Effectiveness) Monitoring and Control System**. The system is built with a Mitsubishi FX5U PLC programmed in GX Works3 and an operator interface (HMI) designed in GT Designer3 and tested in GT Simulator3.

The report is based on three items prepared during the project: the printed ladder diagram of the program (4 pages, 87 rows, dated 06-11-2026), the project device list (a spreadsheet listing every device, its description and the HMI object that uses it), and five HMI screenshots. Everything stated about the program and the screens in this report is taken from those three sources. Where something could not be confirmed from them, the report says so and lists it as an item to verify rather than presenting it as a fact.

1.2 Aim

To design, program and present an OEE monitoring system for a simulated conveyor production line in which a PLC counts products, tracks run time and down time, detects faults and calculates Availability, Performance, Quality and OEE, while an HMI lets the operator control the machine and view all production data.

1.3 Objectives

- Understand OEE and its three factors: Availability, Performance and Quality.
- Allocate PLC devices (inputs, outputs, internal relays and data registers) for the system.
- Write ladder logic in GX Works3 for initialisation, mode selection, start/stop, fault handling, counting, timing and OEE calculation.
- Design four operator screens (Home, Production Monitor, Settings/Modes and Alarm) in GT Designer3 and link every screen object to a PLC device.

- Test the PLC program and the HMI together using the GX Works3 and GT Simulator3 simulators.
- Record the project status honestly, including observations that still need to be verified.

1.4 Scope and limitations

The project runs entirely in software simulation. No physical sensor, motor or actuator is connected. The product sensor and reject sensor are represented by test buttons on the HMI, and the lamps, motor and buzzer are PLC outputs that are observed in the simulator. The device list also reserves physical inputs X0 to X4 for a later stage when real sensors are wired, but these are not part of the current ladder program.

2. Background Theory

2.1 Programmable Logic Controller (PLC)

A PLC is an industrial computer designed to control machines and processes. It reads inputs (sensors, buttons), executes a stored program and writes outputs (motors, lamps, valves). The program is executed repeatedly in a loop called the **scan cycle**: the PLC reads the input states, solves the ladder program from the first rung to the END instruction, and then updates the outputs. This scan order matters in this project because several rungs reuse the same scratch registers one after another within a single scan.

2.2 Mitsubishi FX5U and GX Works3

The FX5U is a compact PLC from the Mitsubishi MELSEC iQ-F family. **GX Works3** is the engineering software used to write, compile, monitor and debug the program. It contains a simulator (GX Simulator3) that behaves like a PLC CPU so the program can be run and monitored without hardware. The program for this project is a single ladder program (data name ProgPou) ending with an END instruction at step 485.

2.3 Ladder logic

Ladder logic is a graphical language in which a rung is read from left to right. Contacts on the left are conditions; the instruction or coil on the right is the action. Contacts in series behave as AND, contacts in parallel behave as OR, and a normally closed contact (drawn with a diagonal bar) is true when its device is OFF. The ladder in this project is printed with 12 columns, so most conditions sit in columns 1 to 8 and the actions (coils, SET, RST and application instructions) sit in columns 9 to 12.

2.4 Instructions used in the program

Table 1: Ladder instructions used in this project

Instruction	Meaning (as used here)
Contact / coil	A contact tests a bit device. A coil (for example Y0 or M200) is ON while the rung condition is true and OFF otherwise.
SET / RST	SET latches a bit ON and it stays ON; RST resets (turns OFF) a bit. Used for the fault latch M201, the run latch M205 and mode bits.
Rising-edge contact	Is true for one scan when the device changes from OFF to ON. Used on M105 and M106 so one button press is one count.
PLS	Turns its output ON for one scan on the rising edge of the rung condition. Used to create M122, M121 and M220.

Instruction	Meaning (as used here)
MOV	Copies one value or register into another, for example `MOV D124 D108`.
FMOV	Fills a block of consecutive registers with one value. `FMOV K0 D100 K6` writes 0 into D100 to D105.
INC	Adds 1 to a register each time it is executed. It must be combined with a one-scan condition or it would add 1 every scan.
SUB	Subtraction. `SUB D100 D102 D101` calculates D101 = D100 - D102.
MUL / DMUL	MUL multiplies two 16-bit values into a 32-bit result (two registers). DMUL multiplies two 32-bit values into a 64-bit result (four registers).
DDIV	Divides a 32-bit value by a 32-bit value. The quotient is stored in two registers and the remainder in the next two. The result is a whole number (any fraction is dropped).
Comparison contacts	>, >= and <= contacts compare two values and conduct when the comparison is true, for example D103 > K0.
SM400 / SM402 / SM412	Special relays: SM400 is always ON, SM402 is ON for one scan when the PLC enters RUN, and SM412 is a 1-second clock. The device list gives the older FX-series names as M8000, M8002 and M8013.

A 32-bit value occupies two consecutive registers. For example the 32-bit pair D120 and D121 is referred to as D120, and a DDIV that uses D124 writes its quotient into D124 and D125 and its remainder into D126 and D127. This is why the OEE calculation uses a block of scratch registers D120 to D135 that the device list marks as "do not reuse".

2.5 Human-Machine Interface (HMI)

An HMI is the screen through which an operator controls and monitors a machine. In this project a Mitsubishi GOT is simulated. **GT Designer3** is used to draw screens and place objects such as switches, lamps, numerical displays and an alarm viewer. Each object is linked to a PLC device. **GT Simulator3** runs the screen project on the computer; its status bar in the screenshots shows that it is working with GX Simulator3, which is how the screens read and write the PLC devices without physical hardware.

2.6 Types of HMI objects used

Table 2: HMI object types seen in the screenshots

Object	Function in this project
Lamp	Shows the state of a bit device, for example RUNNING (M200), AUTO (M202), MANUAL (M203), FAULT (M201).
Switch / button	Writes to a bit device. The device list describes the command buttons as momentary: the bit is ON only while the button is pressed.
Numerical display	Shows the value of a data register, for example D100 total production.
Numerical input	Lets the operator type a value into a data register. Used for Ideal Cycle Time (D106) and Target qty (D107). Shown with a yellow background on Screen 3.
Alarm viewer	A table that lists alarm events with occurrence time, comment, recovery and check columns, with buttons to move the cursor, check, delete, save and reset.
Screen-change button	BACK, NEXT and the three Home buttons change the displayed screen.
Clock / date	Time and date objects shown in the header and footer of each screen.

3. Overall Equipment Effectiveness (OEE)

3.1 What is OEE?

OEE is a standard industrial measure of how effectively a machine or production line is used. It combines three losses into one percentage: the machine not running when it should (Availability), the machine running slower than its ideal speed (Performance) and the machine making defective products (Quality). The concept is associated with Seiichi Nakajima and the Total Productive Maintenance (TPM) approach.

3.2 The three factors

Table 3: OEE factors

Factor	Question it answers	Formula
Availability	How much of the planned production time was the machine actually running?	Availability = Runtime / Planned Production Time x 100
Performance	Was the machine producing at its ideal speed while it ran?	Performance = (Ideal Cycle Time x Total Production) / Runtime x 100
Quality	What share of the products was good?	Quality = Good Products / Total Production x 100
OEE	Overall effectiveness	OEE = Availability x Performance x Quality

3.3 Scaling used in the PLC (per-mille)

A PLC register holds whole numbers, so a percentage such as 87.5 % cannot be stored directly. The project therefore stores every result **multiplied by 10** (a per-mille value). The device list gives the example 855 = 85.5 %. The HMI must show the register with one decimal place to present it as a percentage. Because the values are scaled by 10, the formulas in the ladder multiply by 1000 instead of 100 where a ratio is converted, and the OEE product is divided by 1,000,000 (see Section 7).

3.4 Illustrative worked example

The following numbers are an illustration of the arithmetic only. They are not measured results from the simulator. The calculation follows the same integer arithmetic as the ladder, where each division drops the fraction.

Table 4: Illustrative OEE calculation (not measured data)

Quantity	Calculation	Result
Planned time D103	given	600 s
Run time D104	given	525 s
Ideal cycle time D106	given	5 s
Total production D100	given	100 pcs
Reject products D102	given	5 pcs
Good products D101	100 - 5	95 pcs
Availability D108	525 x 1000 / 600 = 875	875 = 87.5 %
Performance D109	(5 x 100 x 1000) / 525 = 952.4, whole number 952	952 = 95.2 %
Quality D110	95 x 1000 / 100 = 950	950 = 95.0 %
OEE D111	875 x 952 x 950 / 1,000,000 = 791.35, whole number 791	791 = 79.1 %

The project notes also use the example Availability 87.5 %, Performance 79.4 % and Quality 95.0 %, which multiply to an OEE of about 66.0 %. The same relationship holds: the OEE is always lower than or equal to the smallest of the three factors.

4. System Description

4.1 Hardware and software

Table 5: Components used

Component	Description
PLC	Mitsubishi FX5U CPU
PLC programming software	GX Works3
PLC simulation	GX Works3 simulator (GX Simulator3)
HMI design software	GT Designer3
HMI simulation	GT Simulator3 (the screenshots show a GS21 GOT project)
Communication	Mitsubishi PLC to GOT communication, simulated between the two simulators
System type	Industrial production monitoring and control (conveyor line)

4.2 System architecture

Figure 1 shows how the parts of the system relate. The operator works only with the HMI. The HMI reads and writes PLC devices. The PLC program decides what the outputs do and calculates the OEE values, which the HMI reads back for display. The product and reject sensors are currently replaced by HMI test buttons that set M105 and M106.

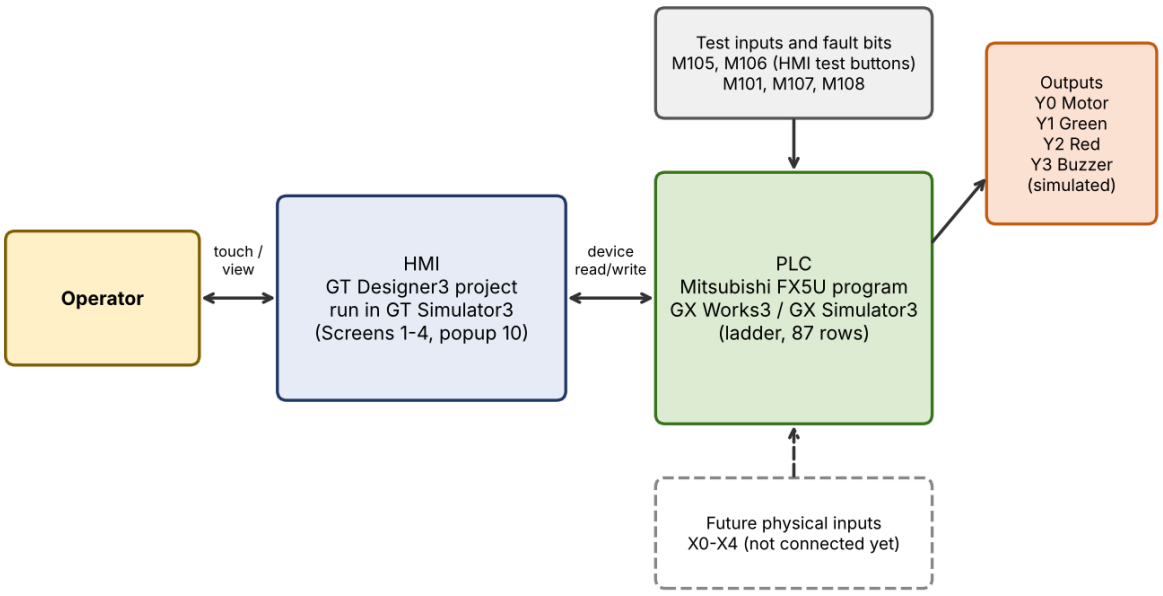
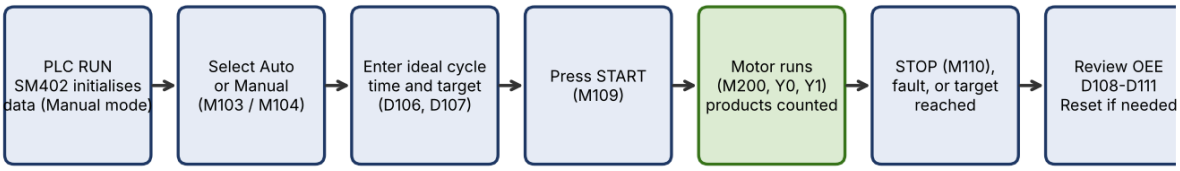


Figure 1: Block diagram of the simulated OEE system

4.3 Operating sequence

The sequence below is the one described in the project brief, mapped onto the final ladder program and device list.

1. The PLC enters RUN. The first-scan relay SM402 initialises the data (Section 6.1). The machine starts in **Manual** mode.
2. The operator selects Auto (M103) or Manual (M104) on the Settings screen. A mode can only be changed while the machine is not running and production is not active.
3. The operator enters the Ideal Cycle Time (D106) and the Target quantity (D107) on the Production Monitor screen.
4. The operator presses START (M109). The PLC checks that no fault is latched and that Stop is not pressed. In Auto mode it also checks that the target has not already been reached.
5. If starting is permitted the run latch M205 is set, the motor-running bit M200 turns ON and the outputs Y0 (motor) and Y1 (green lamp) turn ON.
6. In Auto mode the production-active bit M204 is also set. While the machine runs and M204 is ON, each press of the Product test button (M105) adds 1 to total production D100 and each press of the Reject test button (M106) adds 1 to the reject count D102.
7. Planned time, run time and down time are counted in seconds from the 1-second pulse M220 while M204 is ON.
8. A fault condition (E-stop M101, Fault input M107 or M108) sets the fault latch M201, stops the run latch, turns ON the red lamp Y2 and makes the buzzer Y3 blink.
9. The operator stops with STOP (M110), or the machine stops by itself when the target is reached in Auto mode.
10. The PLC continuously calculates Availability, Performance, Quality and OEE and the HMI displays them.



Fault (M213/M201) stops the run latch; it is cleared with Clear Fault (M111) only when no fault condition remains

Figure 2: Operating flow of the production line

4.4 Operating states

Table 6: Machine states derived from the ladder logic

State	Bits	Visible indication
Machine ready	M200 OFF and M201 OFF (M212 ON)	MACHINE READY lamp on Screen 1
Running	M200 ON (Y0 and Y1 ON)	RUNNING lamp; green lamp output Y1
Fault	M201 ON (Y2 ON, Y3 blinking)	FAULT lamp; alarm entry on Screen 4
Auto mode active	M202 ON	AUTO lamp
Manual mode active	M203 ON	MANUAL lamp
Production active	M204 ON (Auto mode only)	Counters and timers update
Target reached	M211 ON	In Auto mode the run latch is reset

5. PLC Device Allocation

This section reproduces the project device list. It is the final reference mapping and replaces the provisional mapping used at the start of the project (for example, Start was M100 and the buzzer was Y5 in the earlier notes; the final program uses Start = M109, Stop = M110 and Buzzer = Y3). Devices are grouped by category.

5.1 Physical inputs (reserved for later)

Table 7: Physical inputs

Device	Description	HMI object / screen	Notes
X0	Product sensor	Not on an HMI screen	Replaces M105 when wired
X1	Reject sensor	Not on an HMI screen	Replaces M106 when wired
X2	E-stop	Not on an HMI screen	Replaces M101 when wired
X3	Fault input 1	Not on an HMI screen	Replaces M107 when wired
X4	Fault input 2	Not on an HMI screen	Replaces M108 when wired

5.2 Outputs

Table 8: PLC outputs

Device	Description	HMI object / screen	Notes
Y0	Motor	Not on an HMI screen	
Y1	Green lamp (running)	Not on an HMI screen	
Y2	Red lamp (fault)	Not on an HMI screen	
Y3	Buzzer (blinks on fault)	Not on an HMI screen	

5.3 HMI command bits

Table 9: HMI command bits

Device	Description	HMI object / screen	Notes
M102	Reset Counters	Screen 3 - Reset Counters (momentary)	
M103	Auto mode select	Screen 1 - Auto Mode (momentary)	
M104	Manual mode select	Screen 1 - Manual Mode (momentary)	
M105	Product sensor (test)	Screen 2 - Product button (momentary)	Temporary; remove when real sensor wired
M106	Reject sensor (test)	Screen 2 - Reject button (momentary)	Temporary; remove when real sensor wired
M109	Start	Screen 3 - Start (momentary)	
M110	Stop	Screen 3 - Stop (momentary)	
M111	Clear Fault	Screen 3 / 4 - Clear Fault (momentary)	

Device	Description	HMI object / screen	Notes
M112	Reset System	Screen 3 - Reset System (momentary, via popup)	

5.4 Alarm and fault bits

Table 10: Alarm and fault bits

Device	Description	HMI object / screen	Notes
M101	E-stop	Alarm list	
M107	Fault input 1	Alarm list	
M108	Fault input 2	Alarm list	
M213	Any fault condition	Not on an HMI screen	
M201	Fault latch (alarm active)	Screen 2 Fault lamp; Screen 4 alarm text	

5.5 Status bits

Table 11: Status bits

Device	Description	HMI object / screen	Notes
M200	Motor running	Screen 1 / 2 - Running lamp	
M202	Auto mode	Screen 2 / 3 - Auto lamp	
M203	Manual mode	Screen 2 / 3 - Manual lamp	
M204	Production active	Not on an HMI screen	
M205	Run latch	Not on an HMI screen	
M211	Target reached	Not on an HMI screen	
M212	Machine ready	Screen 1 - Machine Ready lamp	

5.6 Internal pulses and special relays

Table 12: Internal pulses

Device	Description	HMI object / screen	Notes
M121	Counter reset pulse	Not on an HMI screen	
M122	System reset pulse	Not on an HMI screen	
M220	1-second tick	Not on an HMI screen	

Table 13: Special relays

Device	Description	HMI object / screen	Notes
SM402	First scan	Not on an HMI screen	FX series: M8002
SM400	Always ON	Not on an HMI screen	FX series: M8000
SM412	1 s clock	Not on an HMI screen	FX series: M8013

5.7 Data registers

Table 14: Data registers

Device	Description	HMI object / screen	Notes
D100	Total production (pcs)	Screen 2 - Display	
D101	Good products (pcs)	Screen 2 - Display	
D102	Reject products (pcs)	Screen 2 - Display	
D103	Planned time (s)	Screen 2 - Display	
D104	Run time (s)	Screen 2 - Display	
D105	Down time (s)	Screen 2 - Display	
D106	Ideal cycle time (s)	Screen 2 - Input	
D107	Target quantity (pcs)	Screen 2 - Input	
D108	Availability (per-mille)	Screen 2 - Display	855 = 85.5 %
D109	Performance (per-mille)	Screen 2 - Display	855 = 85.5 %
D110	Quality (per-mille)	Screen 2 - Display	855 = 85.5 %
D111	OEE (per-mille)	Screen 2 - Display	855 = 85.5 %

Table 15: Scratch registers

Device	Description	HMI object / screen	Notes
D120-D135	OEE math workspace	Not on an HMI screen	32-bit pairs for DMUL/DDIV; do not reuse

5.8 Devices marked not used

The device list marks the following devices as left over from an earlier version of the program that should be removed. They do not appear in the final ladder printout.

Table 16: Devices marked for deletion

Device	Description	HMI object / screen	Notes
M100	Old motor latch	Not on an HMI screen	Remove from old program
M120	Old reset pulse	Not on an HMI screen	Remove from old program
M207	Old blink flag	Not on an HMI screen	Remove from old program
M208	Old count pulse	Not on an HMI screen	Remove from old program
M209	Old reject pulse	Not on an HMI screen	Remove from old program
T0 / T1	Old timers	Not on an HMI screen	Remove from old program

5.9 HMI screens

Table 17: HMI screen numbers

Device	Description	HMI object / screen	Notes
1	Home	Not on an HMI screen	
2	Production Monitor	Not on an HMI screen	
3	Settings / Modes	Not on an HMI screen	
4	Alarm / Fault	Not on an HMI screen	
10	Reset confirm popup	Not on an HMI screen	Window screen

5.10 Test inputs versus real inputs

The device list shows the planned replacement when real hardware is wired: X0 replaces the Product test bit M105, X1 replaces the Reject test bit M106, X2 replaces the E-stop bit M101, and X3 and X4 replace the fault bits M107 and M108. Until then M105 and M106 are marked "Temporary; remove when real sensor wired".

6. Ladder Program Description

The ladder program has 87 rows over four printed pages and ends with END at step 485. In the printout each rung starts with its step number in brackets, for example (28). The program is described here rung by rung, grouped by function. The number after each heading is the step number where the rung starts and the row numbers refer to the printed ladder pages shown in Figures 3 to 6.

6.1 Initialisation, mode selection and faults (page 1)

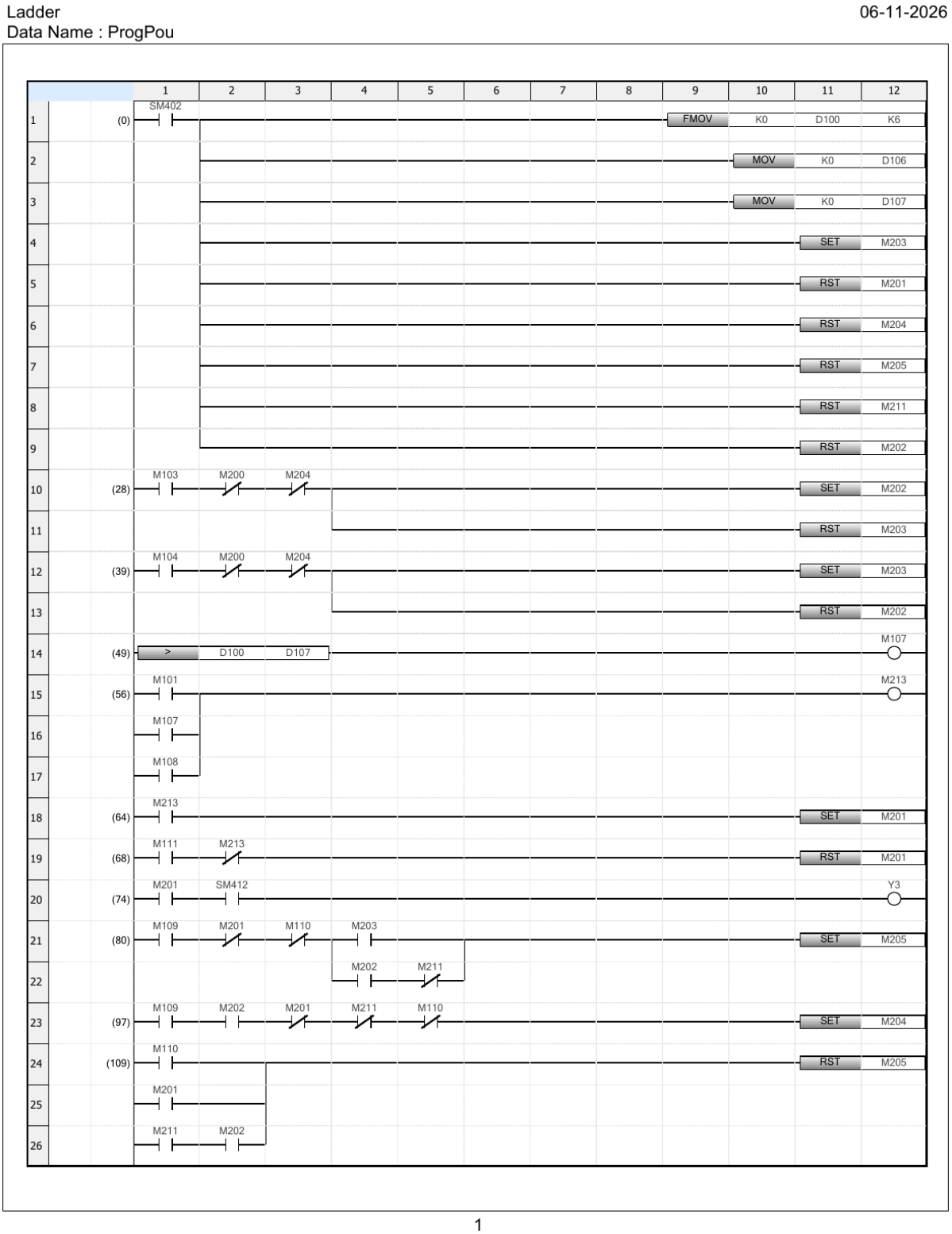


Figure 3: Ladder diagram, page 1 (rows 1 to 26)

Rung at step 0, rows 1 to 9: Initialisation

When SM402 is ON for the first scan after RUN, the program writes 0 into D100 to D105 using FMOV (total, good, reject, planned time, run time and down time), and writes 0 into D106 (ideal cycle time) and D107 (target). It then sets M203 (Manual mode ON) and resets M201 (fault latch), M204 (production active), M205 (run latch), M211 (target reached) and M202 (Auto mode).

Why it is needed: the machine must start from a known state every time the PLC enters RUN. SM402 is used instead of the always-ON SM400, which would clear the counters on every scan.

> **Observation:** Because D106 and D107 are cleared at the first scan, the Ideal Cycle Time and Target qty must be entered again on the HMI after each PLC start.

Rungs at steps 28 and 39, rows 10 to 13: Mode selection

M103 with M200 OFF and M204 OFF sets M202 (Auto) and resets M203 (Manual). M104 with M200 OFF and M204 OFF does the opposite. Because each rung sets one bit and resets the other, Auto and Manual are mutually exclusive. The conditions M200 OFF and M204 OFF prevent a mode change while the machine is running or while production is active.

Rung at step 49, row 14: Target comparison

A comparison contact tests D100 > D107 (total production greater than the target) and drives the coil M107. In the device list M107 is described as "Fault input 1" and it appears in the alarm list.

Rung at step 56, rows 15 to 17: Any fault condition

M101 (E-stop), M107 (Fault input 1) and M108 (Fault input 2) are in parallel and drive the coil M213, "any fault condition". If any of the three is ON, M213 is ON.

Rungs at steps 64 and 68, rows 18 and 19: Fault latch and clear

M213 sets the fault latch M201. M111 (Clear Fault) resets M201 only while M213 is OFF. Therefore pressing Clear Fault does not clear a fault that is still present, and clearing a fault does not itself restart the machine (the machine must be started again with START).

Rung at step 74, row 20: Buzzer

M201 in series with the 1-second clock SM412 drives Y3. While a fault is latched, the buzzer output follows the clock and therefore blinks once per second.

Rung at step 80, rows 21 and 22: Run latch set

The run latch M205 is set when M109 (Start) is pressed, M201 is OFF (no fault latched), M110 (Stop) is OFF and one of the following is true: Manual mode is active (M203), or Auto mode is active (M202) and the target is not reached (M211 OFF).

Rung at step 97, row 23: Production active set

The production-active bit M204 is set when M109 is pressed in Auto mode (M202) with no fault latched, the target not reached and Stop not pressed. Only Auto mode sets M204.

Rungs at steps 109 and 120, rows 24 to 29: Run latch and production reset

The run latch M205 is reset by M110 (Stop), or M201 (fault latched), or M211 together with M202 (target reached in Auto mode). Production-active M204 is reset by M110, M211 or M203. The last condition means that M204 can never stay ON in Manual mode.

6.2 System reset, outputs and counting (page 2)

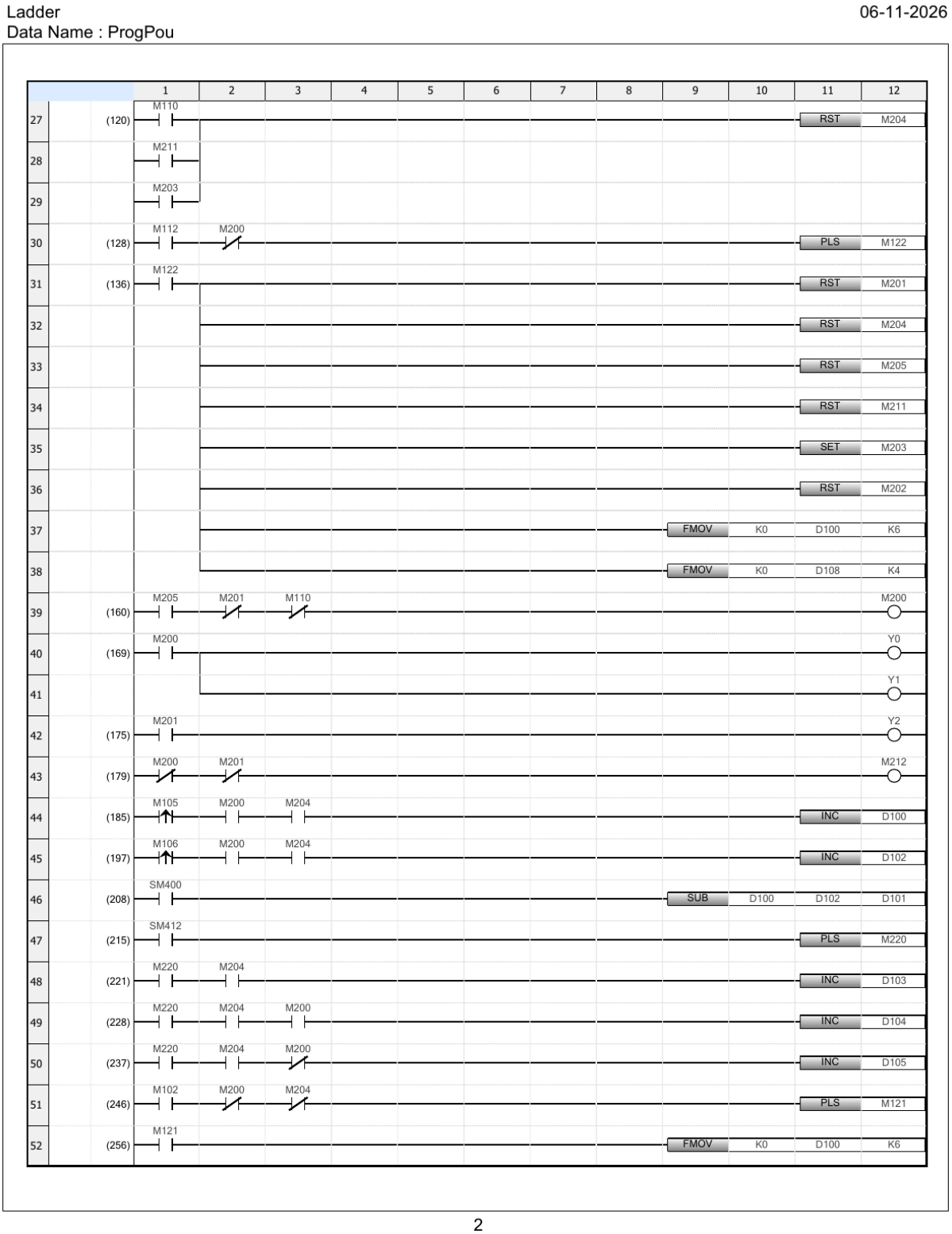


Figure 4: Ladder diagram, page 2 (rows 27 to 52)

Rungs at steps 128 and 136, rows 30 to 38: System reset

M112 (Reset System) with M200 OFF creates a one-scan pulse M122 using PLS. On that pulse the program resets M201, M204, M205 and M211, sets Manual mode (sets M203 and resets M202), clears D100 to D105 and clears D108 to D111 (FMOV K0 D108 K4). The ideal cycle time

D106 and the target D107 are not cleared by this reset. The device list states the Reset System button opens through a popup (screen 10).

Rung at step 160, row 39: Machine running

M200 is ON when the run latch M205 is ON, no fault is latched (M201 OFF) and Stop (M110) is not pressed. M200 is the master "motor running" condition for the whole program.

Rungs at steps 169 to 179, rows 40 to 43: Outputs and Machine Ready

M200 drives Y0 (motor) and Y1 (green lamp). M201 drives Y2 (red lamp). M212 (Machine Ready) is ON when both M200 and M201 are OFF, i.e. the machine is neither running nor faulted.

Rungs at steps 185, 197 and 208, rows 44 to 46: Counting

A rising edge on M105 (Product test) increments total production D100 once, but only while M200 and M204 are both ON. A rising edge on M106 (Reject test) increments the reject count D102 once under the same two conditions. A third rung, driven by the always-ON SM400, calculates Good = Total - Reject (D101 = D100 - D102).

> **Observation:** Because the reject button increments only D102, and Good is calculated as D100 - D102, the program assumes that every rejected product is also counted as a product with the Product button. If Reject is pressed without a matching Product press, D101 would become negative.

Rungs at steps 215 to 237, rows 47 to 50: Time counting

SM412 (1-second clock) creates a one-scan pulse M220 every second using PLS. While M204 is ON, each M220 pulse adds 1 to the planned time D103. If M200 is also ON it adds 1 to the run time D104; if M200 is OFF it adds 1 to the down time D105. All three registers therefore count in seconds, and while M204 is ON, planned time always equals run time plus down time.

Rungs at steps 246 and 256, rows 51 and 52: Reset counters

M102 (Reset Counters) with M200 OFF and M204 OFF creates a pulse M121. On that pulse FMOV clears D100 to D105. The OEE result registers are not cleared here, but they are recalculated every scan and are forced to 0 by the protection rungs when the inputs are 0 (Section 7.6).

6.3 Target and OEE calculation (pages 3 and 4)

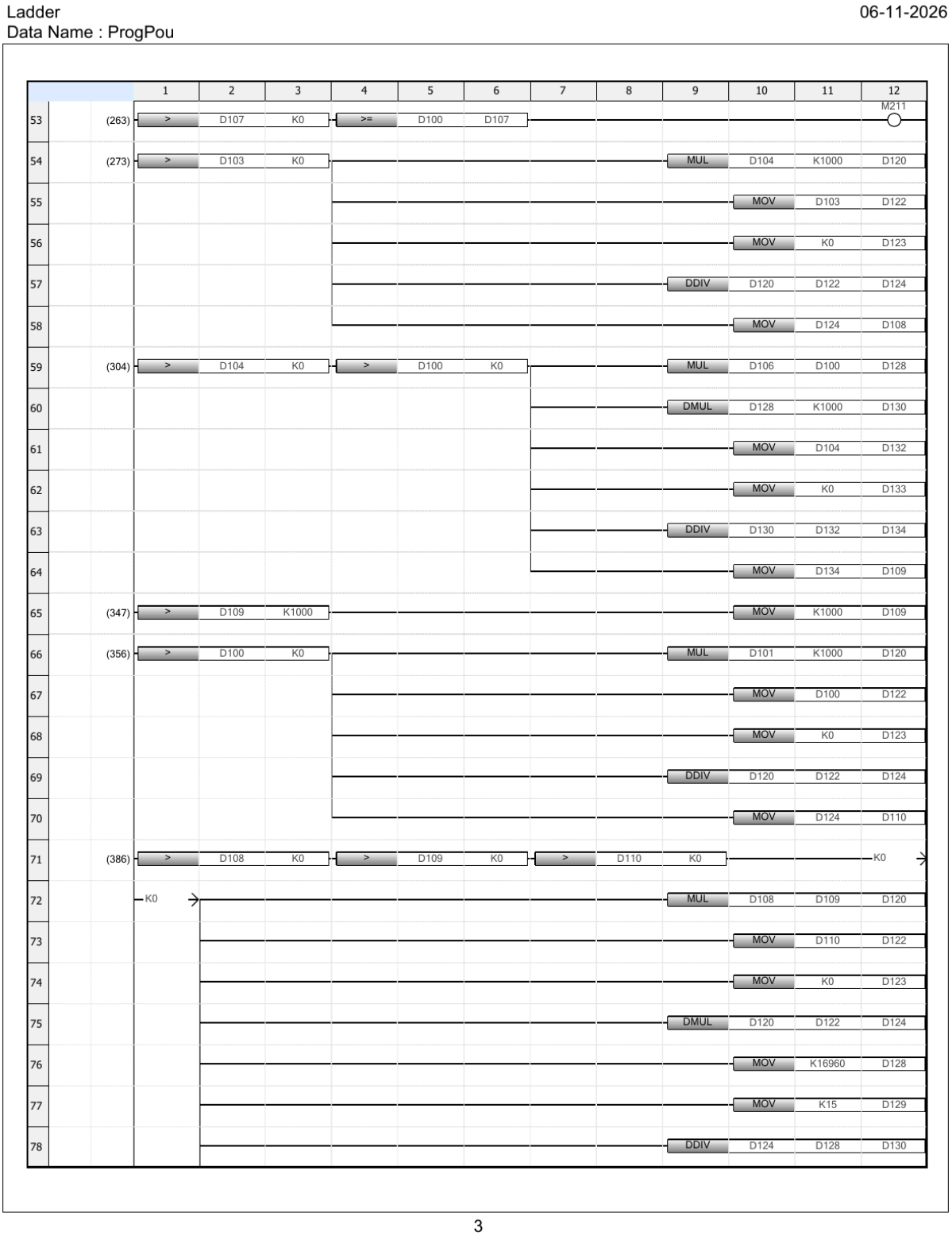


Figure 5: Ladder diagram, page 3 (rows 53 to 78)

Rung at step 263, row 53: Target reached

If the target D107 is greater than 0 and total production D100 is greater than or equal to D107, the coil M211 turns ON. In Auto mode this resets the run latch (Section 6.1), so the machine stops when the target is reached.

The remaining rungs on pages 3 and 4 calculate the OEE and are explained in Section 7.

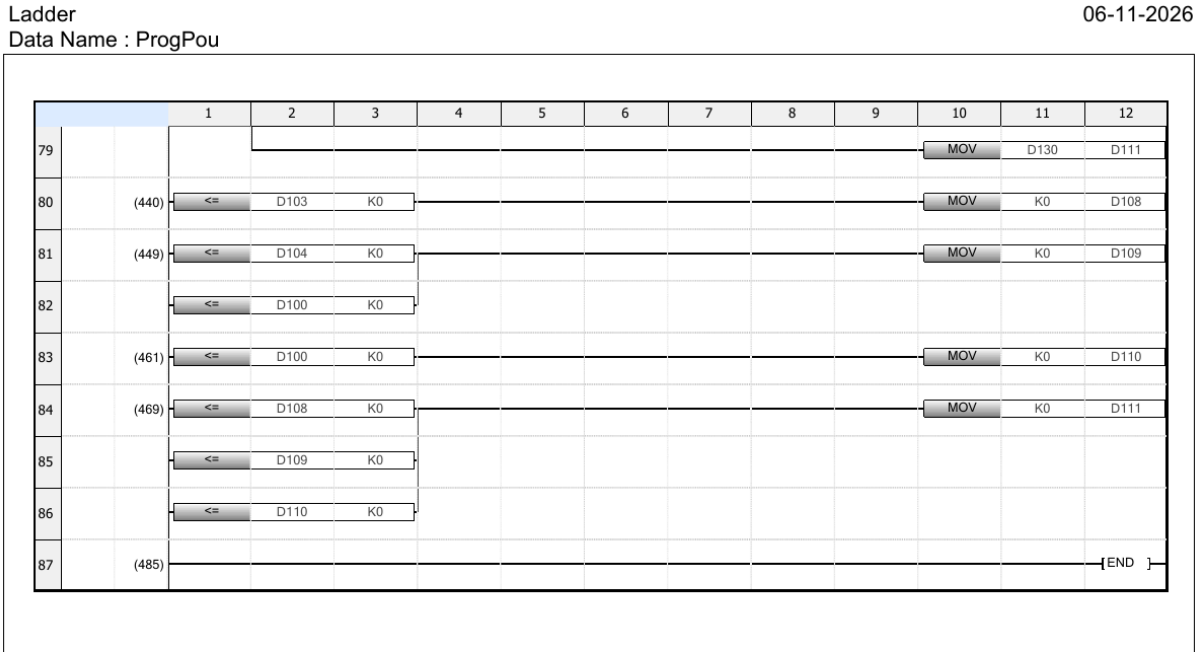


Figure 6: Ladder diagram, page 4 (rows 79 to 87)

6.4 Summary of all rungs

Table 18: Rung summary (step number, rows, function)

Step	Rows	Function	Result
0	1-9	Initialisation (SM402)	Clears D100-D107, sets Manual mode, resets M201, M202, M204, M205, M211
28	10-11	Select Auto	SET M202, RST M203 (needs M200 and M204 OFF)
39	12-13	Select Manual	SET M203, RST M202 (needs M200 and M204 OFF)
49	14	D100 > D107	Coil M107
56	15-17	Any fault	M101 or M107 or M108 drives M213
64	18	Fault latch set	M213 sets M201
68	19	Fault clear	M111 with M213 OFF resets M201
74	20	Buzzer	M201 and SM412 drive Y3
80	21-22	Run latch set	M109 sets M205 under the start conditions
97	23	Production active set	M109 sets M204 in Auto mode
109	24-26	Run latch reset	M110, M201 or (M211 and M202) reset M205
120	27-29	Production active reset	M110, M211 or M203 reset M204
128	30	System reset pulse	M112 and M200 OFF give PLS M122

Step	Rows	Function	Result
136	31-38	System reset actions	Resets bits, sets Manual, clears D100-D105 and D108-D111
160	39	Machine running	M200 = M205 and not M201 and not M110
169	40-41	Motor and green lamp	Y0 and Y1 follow M200
175	42	Red lamp	Y2 follows M201
179	43	Machine ready	M212 = not M200 and not M201
185	44	Product count	Rising edge M105 with M200 and M204 increments D100
197	45	Reject count	Rising edge M106 with M200 and M204 increments D102
208	46	Good count	D101 = D100 - D102
215	47	1-second pulse	SM412 gives PLS M220
221	48	Planned time	M220 and M204 increment D103
228	49	Run time	M220, M204 and M200 increment D104
237	50	Down time	M220 and M204 with M200 OFF increment D105
246	51	Counter reset pulse	M102 with M200 and M204 OFF gives PLS M121
256	52	Counter clear	M121 clears D100-D105
263	53	Target reached	D107 > 0 and D100 >= D107 drive M211
273	54-58	Availability	D108 = D104 x 1000 / D103
304	59-64	Performance	D109 = D106 x D100 x 1000 / D104
347	65	Performance limit	If D109 > 1000 then D109 = 1000
356	66-70	Quality	D110 = D101 x 1000 / D100
386	71-79	OEE	D111 = D108 x D109 x D110 / 1,000,000
440-469	80-86	Zero protection	Forces D108-D111 to 0 when inputs are 0 or less
485	87	END	End of program

7. OEE Calculation in the PLC

7.1 Overview

The calculation is performed every scan using 32-bit arithmetic so that intermediate results do not overflow a 16-bit register (whose limit is 32,767). Each calculation multiplies first and divides last, so that the integer division keeps as much precision as possible. Figure 7 shows the data flow.

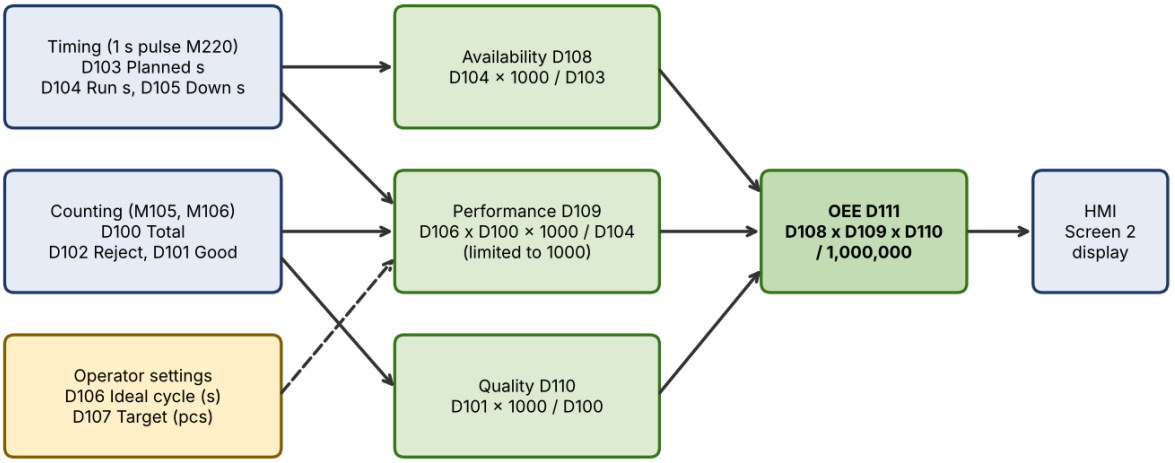


Figure 7: Data flow of the OEE calculation

7.2 Availability (steps 273 to 304)

The rung runs only if the planned time D103 is greater than 0, which prevents division by zero. The steps are:

1. `MUL D104 K1000 D120`: Run time x 1000 is stored as a 32-bit value in D120 and D121.
2. `MOV D103 D122` and `MOV K0 D123`: the planned time is placed in D122 and the upper word D123 is cleared, forming a 32-bit divisor.
3. `DDIV D120 D122 D124`: the 32-bit dividend is divided by the 32-bit divisor. The quotient goes to D124 and D125.
4. `MOV D124 D108`: the quotient is copied to D108 (Availability x 10).

7.3 Performance (steps 304 to 356)

The rung runs only if run time D104 > 0 and total production D100 > 0.

1. `MUL D106 D100 D128`: Ideal cycle time x Total production (the ideal time needed to make the products), as a 32-bit value in D128 and D129.
2. `DMUL D128 K1000 D130`: multiplied by 1000. The 64-bit result occupies D130 to D133.
3. `MOV D104 D132` and `MOV K0 D133`: the run time becomes the 32-bit divisor in D132 and D133.
4. `DDIV D130 D132 D134`: divides the value in D130 (lower 32 bits) by the run time. The quotient is in D134 and D135.

5. `MOV D134 D109`: the result is stored in D109 (Performance x 10).
6. Step 347: if D109 > 1000 it is limited to 1000 (100.0 %).

> **Observation:** The DMUL result takes four registers (D130 to D133) and the program then reuses D132 and D133 as the divisor. This works while the product (ideal cycle time x total production x 1000) fits in 32 bits, i.e. is below 2,147,483,647. For the quantities used in this project this is far from the limit (for example, 5 s x 100 pcs x 1000 = 500,000), but it is a design limit to be aware of for very large counts.

7.4 Quality (steps 356 to 386)

The rung runs only if total production D100 > 0. `MUL D101 K1000 D120`, `MOV D100 D122`, `MOV K0 D123`, `DDIV D120 D122 D124` and `MOV D124 D110` calculate Good x 1000 / Total and store it in D110 (Quality x 10).

7.5 Overall OEE (steps 386 to 440)

The rung runs only if D108, D109 and D110 are all greater than 0.

1. `MUL D108 D109 D120`: Availability x Performance (each scaled by 10) as a 32-bit value.
2. `MOV D110 D122` and `MOV K0 D123`: Quality is prepared as a 32-bit value.
3. `DMUL D120 D122 D124`: the three scaled values are multiplied. The 64-bit result occupies D124 to D127.
4. `MOV K16960 D128` and `MOV K15 D129`: the 32-bit constant 1,000,000 is built from two words (15 x 65,536 + 16,960 = 1,000,000).
5. `DDIV D124 D128 D130`: the product is divided by 1,000,000. The quotient is in D130 and D131.
6. `MOV D130 D111`: the result is stored in D111 (OEE x 10).

The division by 1,000,000 is correct for per-mille values. Each stored factor equals 1000 x its fraction (for example 875 means 0.875), so the product of the three is 1,000,000,000 x (A x P x Q as fractions). Dividing by 1,000,000 leaves 1000 x OEE as a fraction, which is the OEE per-mille value. For example 875 x 952 x 950 / 1,000,000 = 791, which means 79.1 %. The largest possible product is 1000 x 1000 x 1000 = 1,000,000,000, which is below the 32-bit limit, so the DDIV input does not overflow.

7.6 Protection against zero values

Table 19: Zero-protection rungs

Step	Rows	Condition	Action
440	80	Planned time D103 <= 0	D108 = 0
449	81-82	Run time D104 <= 0 or Total D100 <= 0	D109 = 0
461	83	Total production D100 <= 0	D110 = 0
469	84-86	D108 <= 0 or D109 <= 0 or D110 <= 0	D111 = 0

These rungs are the counterpart of the calculation rungs: whenever a calculation is skipped because its divisor would be zero, the matching result register is forced to 0 so the HMI never shows an old value. Together with the conditions on the calculation rungs, they protect the program against division by zero.

7.7 Register summary

Table 20: Registers used by the OEE calculation

Register	Contents	Unit / scaling
D100	Total production	pieces
D101	Good products (D100 - D102)	pieces
D102	Reject products	pieces
D103	Planned time	seconds
D104	Run time	seconds
D105	Down time	seconds
D106	Ideal cycle time (operator input)	seconds
D107	Target quantity (operator input)	pieces
D108 to D111	Availability, Performance, Quality, OEE	per-mille (855 = 85.5 %)
D120 to D135	Scratch registers for 32/64-bit maths	internal use only

8. HMI Design

8.1 Design approach

The project brief asked for a professional, minimalist industrial interface with clear text, green for normal operation, red for faults, amber or yellow for warnings and grey for inactive items. The implemented screens use a consistent layout: a dark blue header bar with the Mitsubishi logo, screen title and clock; a white rounded panel for the content; and a dark blue footer bar with navigation buttons and the date. The header and footer colour is navy blue, which differs from the dark grey suggested in the brief, but the layout is the same on every screen. Lamps are dark when OFF and coloured when ON. The START button is green, the STOP button is red and the Reset System button is yellow.

8.2 Screen list

Table 21: HMI screens

Screen	Name	Content
1	Home	Title, welcome message, conveyor picture, navigation buttons, MACHINE READY and RUNNING lamps
2	Production Monitor	Production counts, times, OEE results, settings, and Machine Status page with test buttons
3	Settings / Modes	Mode selection, START and STOP, Reset Counters, Clear Fault, Reset System
4	Alarm / Fault	Alarm viewer table and FAULT lamp
10	Reset confirm popup	Window screen for confirming a reset (listed in the device list; no screenshot supplied)

8.3 Screen 1: Home



Figure 8: Screen 1: Home screen in GT Simulator3

The Home screen shows the title "PRODUCTION LINE CONTROLLER" with the Mitsubishi Electric logo and a clock in the header. The centre shows the heading "WELCOME!", a conveyor illustration and the text "Select an operation mode to begin." Below are three navigation buttons: **MODE Monitor** (gear icon), **Production Monitor** (bar chart icon) and **Alarms** (warning triangle icon). The footer shows the **MACHINE READY** lamp (green in the screenshot, meaning the machine is not running and not faulted) and the **RUNNING** lamp (dark, meaning the machine is not running), and the date. According to the device list, the Machine Ready lamp uses M212 and the Running lamp uses M200.

8.4 Screen 2: Production Monitor

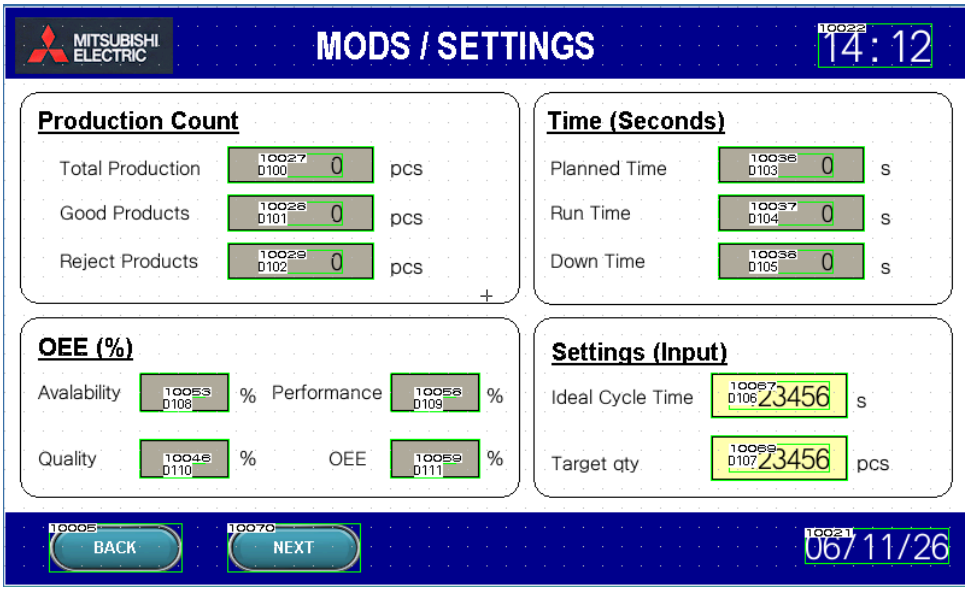


Figure 9: Production Monitor screen with counts, times, OEE and settings

This screen has four panels. The object numbers and devices in the figure are overlaid by the editor.

Table 22: Production Monitor objects

Panel / label	Device	Function
Production Count: Total Production	D100	Numerical display, pieces
Production Count: Good Products	D101	Numerical display, pieces
Production Count: Reject Products	D102	Numerical display, pieces
Time (Seconds): Planned Time	D103	Numerical display, seconds
Time (Seconds): Run Time	D104	Numerical display, seconds
Time (Seconds): Down Time	D105	Numerical display, seconds
OEE (%): Availability	D108	Numerical display, per-mille value shown as %
OEE (%): Performance	D109	Numerical display, per-mille value shown as %
OEE (%): Quality	D110	Numerical display, per-mille value shown as %
OEE (%): OEE	D111	Numerical display, per-mille value shown as %

Panel / label	Device	Function
Settings (Input): Ideal Cycle Time	D106	Numerical input (yellow field), seconds
Settings (Input): Target qty	D107	Numerical input (yellow field), pieces
BACK / NEXT	-	Screen-change buttons

The screenshot is an editor view: the counters show 0 and the two input fields show the placeholder value 23456 (the digits the editor uses to preview a field). The spelling "Avalability" on the label for D108 should be corrected to "Availability".

> **To verify:** The device list says D108 to D111 hold per-mille values (855 = 85.5 %), but the screenshot does not show whether the four OEE fields are configured to show one decimal place. In GT Designer3 the numerical display should be set up so the value is shown with one digit after the decimal point.

8.5 Machine Status page (test buttons)

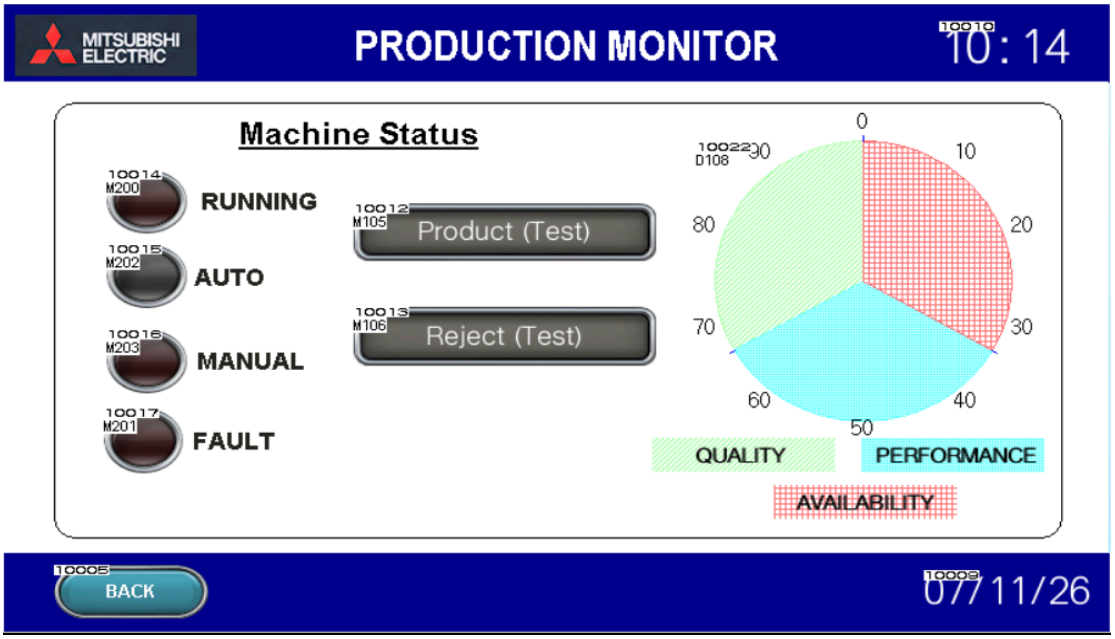


Figure 10: Machine Status page with lamps and test buttons

This page shows four lamps: **RUNNING** (M200), **AUTO** (M202), **MANUAL** (M203) and **FAULT** (M201). On the right are two buttons, **Product (Test)** (M105) and **Reject (Test)** (M106), which simulate the product and reject sensors. A BACK button returns to the previous screen. The device list places the Product and Reject buttons on Screen 2 and describes them as temporary.

8.6 Screen 3: Controls

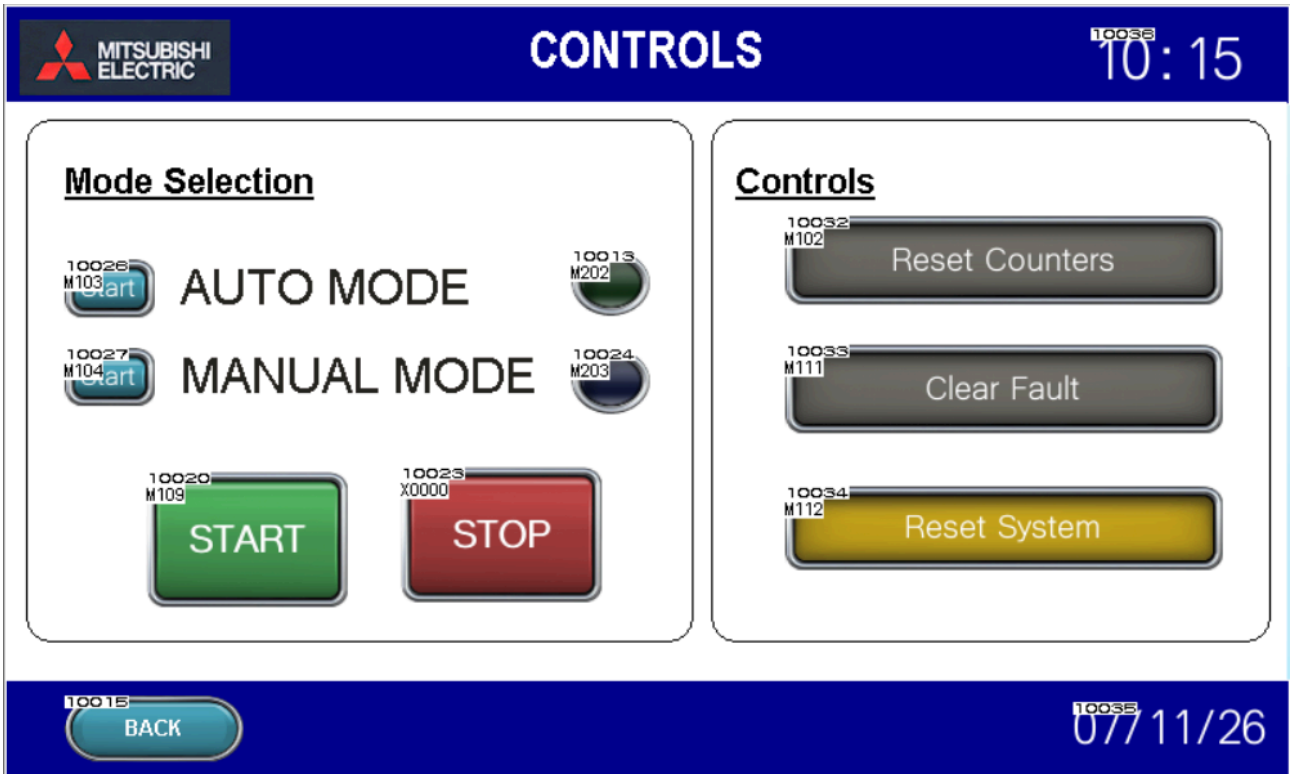


Figure 11. Controls screen with mode selection and controls

Table 23: Settings / Modes objects

Object	Device	Function
AUTO MODE button	M103	Selects Auto mode (momentary)
AUTO lamp	M202	Auto mode active
MANUAL MODE button	M104	Selects Manual mode (momentary)
MANUAL lamp	M203	Manual mode active
START (green)	M109	Requests start (momentary)
STOP (red)	M110	See note below; the ladder and device list use M110
Reset Counters	M102	Clears production counters (D100-D105)
Clear Fault	M111	Clears the fault latch if no fault condition remains
Reset System (yellow)	M112	Full system reset; device list says it is used via popup screen 10
BACK	-	Returns to the previous screen

8.7 Screen 4: Alarm / Fault

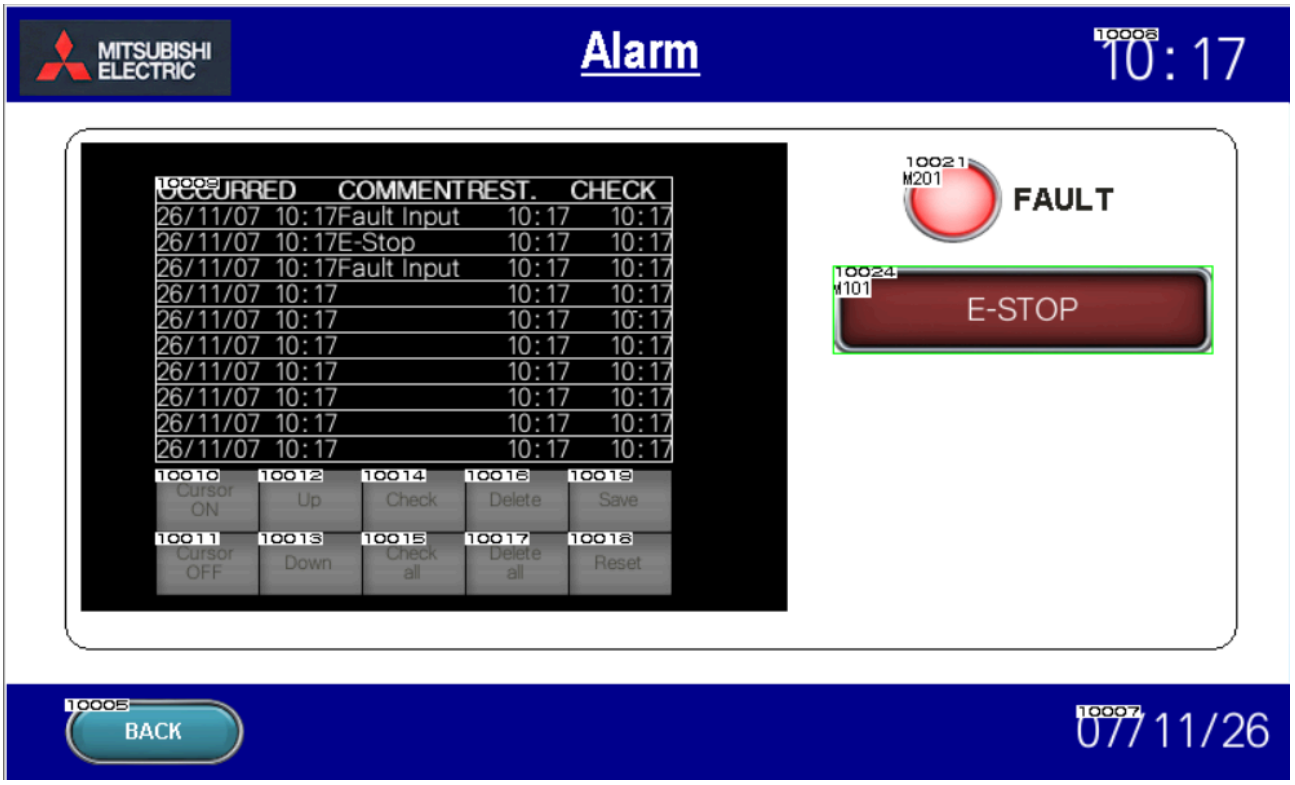


Figure 12: Alarm screen with the alarm viewer and FAULT lamp

The Alarm screen contains an **alarm viewer** with the columns OCCURRED, COMMENT, REST. (recovery) and CHECK, followed by the viewer buttons Cursor ON, Cursor OFF, Up, Down, Check, Check all, Delete, Delete all, Save and Reset. A **FAULT** lamp on the right is linked to M201. The BACK button returns to the previous screen. In the screenshot three alarm entries are listed, with the comments "Fault Input", "E-Stop" and "Fault Input", all time-stamped 26/11/06 14:11. These match the alarm bits in the device list: M101 (E-stop), M107 (Fault input 1) and M108 (Fault input 2).

The device list also assigns a Clear Fault button (M111) to Screens 3 and 4. The Clear Fault button is visible on Screen 3; no such button is visible on the Alarm screen in the screenshot.

8.8 Navigation

The Home screen provides three navigation buttons (MODE Monitor, Production Monitor, Alarms). Every other screen has a BACK button in the bottom-left corner, and the Production Monitor data page also has a NEXT button. The exact target screen of each button is defined in GT Designer3 and cannot be read from the screenshots.

8.9 Documentation consistency

Comparing the screenshots with the device list shows some differences that should be tidied so the documentation matches the project:

- The device list places Auto and Manual mode buttons (M103, M104) on "Screen 1", but the screenshots show them on the Settings / Modes screen; the Home screen shows MODE Monitor, Production Monitor and Alarms buttons.

- The device list describes the Reset System button as opened through a popup (screen 10). The screenshot shows the button on Screen 3, and no screenshot of the popup was supplied.
- The STOP button device (X0000 versus M110), as described in Section 8.6.
- The Production Monitor and Machine Status pages carry the title "MODS / SETTINGS".

9. PLC and HMI Integration

9.1 Data exchange

The HMI does not contain any control logic. It only reads PLC devices (lamps, numerical displays, alarm viewer) and writes PLC devices (buttons and numerical inputs). All decisions, such as whether the machine may start, are taken in the ladder program.

Table 24: Direction of data between HMI and PLC

Direction	Devices	Purpose
HMI writes to PLC (bits)	M102, M103, M104, M105, M106, M109, M110, M111, M112	Operator commands and simulated sensors
HMI writes to PLC (registers)	D106, D107	Ideal cycle time and target quantity
PLC to HMI (bits)	M200, M201, M202, M203, M212	Lamps
PLC to HMI (registers)	D100-D105, D108-D111	Counts, times and OEE results
PLC to HMI (alarms)	M101, M107, M108	Alarm viewer entries

9.2 Why momentary buttons are used

The device list marks all command buttons as momentary. A momentary button turns its bit ON only while it is pressed. This works well with the ladder because the program either uses the bit to set or reset a latch (START, STOP, mode buttons) or converts it to a one-scan pulse (rising-edge contact, PLS). The program therefore does not depend on how long the button is held.

9.3 Example sequence

The following sequence is derived from the ladder logic and shows what is expected when an operator runs a short Auto-mode production.

Table 25: Expected behaviour for an Auto-mode run

Step	Operator action	Expected PLC / HMI response
1	Press AUTO MODE (M103)	M202 ON, M203 OFF; AUTO lamp on
2	Enter D106 and D107 (for example 5 s and 10 pcs)	Values stored in D106 and D107
3	Press START (M109)	M205 and M204 set; M200, Y0, Y1 ON; RUNNING lamp on
4	Press Product (Test) (M105)	D100 increases by 1; D103 and D104 increase every second
5	Press Reject (Test) (M106) after a product	D102 increases by 1; D101 = D100 - D102
6	Press STOP (M110)	M205, M204 reset; M200 OFF; Y0 and Y1 OFF
7	Read Screen 2	D108-D111 show Availability, Performance, Quality and OEE

These are expected results derived from reading the program; they become results only when they are observed in the simulator (see Section 10).

10. Testing

10.1 Test setup

Testing is carried out with the GX Works3 simulator running the ladder program and GT Simulator3 running the HMI project, with the PLC devices monitored in GX Works3. The project is tested in simulation only.

10.2 Status recorded in the project notes

The table below records the testing status noted earlier in the project. It was written before the final ladder in this report and should be re-confirmed against the final program.

Table 26: Test status recorded in the project notes

Test	Recorded status	Remark
GX Works3 compilation and simulator start-up	Passed	
PLC RUN mode and ladder monitoring	Passed	
SM402 start-up initialisation	Passed	
Reset operation (M102)	Passed	
Auto / Manual mode selection	Passed	
Start / Stop logic	Needs further verification	M200 did not turn ON in an earlier start test
Product counting	Not yet completed	
Reject counting	Issue reported earlier	Reject pulse seen but D102 did not increment
Fault simulation	Not fully verified	
Timer operation	Not fully verified	
OEE calculations	Not fully verified	
HMI communication	Needs testing	
Full HMI operation	In development	

> **Note:** In the final ladder the counters only count while both M200 (machine running) and M204 (production active) are ON, and M204 is only set in Auto mode. When testing counting, confirm Auto mode and START first, otherwise the count will not change.

10.3 Test plan for the final program

The following tests check the behaviour described in Section 6. The "Expected result" column is derived from the ladder. The last two columns are left blank so that the actual observation and the Pass/Fail decision can be recorded during testing.

Table 27: Test plan (to be completed during testing)

No.	Action	Expected result	Pass/Fail
T1	Switch PLC to RUN	D100-D107 = 0; M203 ON; M202 OFF; M212 ON	Pass
T2	Press AUTO MODE (M103), machine stopped	M202 ON, M203 OFF	Pass
T3	Press MANUAL MODE (M104), machine stopped	M203 ON, M202 OFF	Pass
T4	Manual mode, press START (M109)	M205, M200, Y0, Y1 ON; M204 stays OFF	Pass
T5	While running press AUTO (M103)	Mode does not change (M200 ON)	Pass
T6	Press STOP (M110)	M200, Y0, Y1 OFF	Pass
T7	Auto mode, D107 = 5, press START	M204 and M200 ON	Pass
T8	Press Product (Test) once	D100 = 1 (one count per press)	Pass
T9	Press Reject (Test) once after T8	D102 = 1, D101 = 0	Pass
T10	Observe time registers for 10 s while running	D103 and D104 increase by about 10	Pass
T11	Trigger a fault while running (as in T13) and wait 10 s	D104 stops; D105 increases by about 10 (M204 stays ON)	Fail
T12	Reach D100 = D107	M211 ON; M205 reset; M200 OFF	Pass
T13	Set M101 (E-stop) while running	M213 and M201 ON; Y2 ON; Y3 blinks; M200 OFF	Pass
T14	Press Clear Fault (M111) with M101 still ON	M201 stays ON	Pass
T15	Turn M101 OFF and press Clear Fault	M201 OFF; machine does not restart by itself	Pass
T16	Set M108 (Fault input 2) while running	Same behaviour as T13	Fail
T17	Press Reset Counters (M102), machine stopped	D100-D105 cleared	Pass
T18	Press Reset System (M112), machine stopped	D100-D105 and D108-D111 cleared; Manual mode set	Pass
T19	Enter D106 = 5, make 10 products (2 rejects) over a known time	D108-D111 match hand calculation using Section 3.4	Pass
T20	Check HMI lamps and values against GX Works3 monitor	HMI and PLC values agree	Pass

Note on T11: The ladder resets M204 on STOP, on target reached and in Manual mode, but a fault resets only the run latch M205. So in the final program down time (D105) accumulates after a fault, not after a normal stop. Confirm this in the simulator.

11. Observations from Ladder and HMI Review

The following points were found by reading the ladder printout, the device list and the screenshots. They are observations to be verified in the simulator; none is a confirmed test failure.

Table 28: Review observations

No.	Observation	Why it matters
1	STOP button on Screen 3 is bound to X0000 in the screenshot; the ladder and device list use M110	The ladder stop rungs would not respond to the HMI button if the binding is not M110
2	M107 is driven by a coil from D100 > D107 (step 49) but is also listed as "Fault input 1" and used in the any-fault rung	With D107 = 0 (the value after power-up) the first product count would make D100 > D107, turning M107 ON and latching a fault. Enter the target before counting. It also means M107 cannot be used as an independent simulated fault input while this coil exists
3	Counting, timing and OEE depend on M204, which is only set in Auto mode	No counts or time accumulate in Manual mode
4	D106 and D107 are cleared by SM402 at start-up	Ideal cycle time and target must be re-entered after each PLC start; with D106 = 0 Performance and OEE are 0
5	Good = Total - Reject assumes each reject is also counted as a product	Pressing Reject alone makes Good negative
6	Start (M109) sets both the run latch and production-active bit; there is no separate Shift Start / Shift End	The original brief described a separate shift start and end (M109 and M110); in the final program M109 is Start and M110 is Stop
7	M101 is used as an E-stop bit but is only a PLC bit	A real emergency stop must not depend on PLC logic (see Section 12)
8	OEE fields show per-mille values	The HMI display must be configured with one decimal place
9	Screen titles and the "Avalability" spelling	Cosmetic, but visible to the operator

12. Safety and Limitations

- **Simulation only.** No real motor, sensor or actuator is controlled. The behaviour of real hardware (contact bounce, sensor delay, motor stop time) is not represented.
- **E-stop.** In this project the E-stop is the PLC bit M101, planned to be replaced by input X2. A real emergency stop must be a hard-wired safety device that removes power from the actuators independently of the PLC program and HMI.
- **Stop and fault behaviour.** The machine stops when STOP is pressed or a fault is latched. Clear Fault does not restart the machine; START must be pressed again.
- **Operator input limits.** The HMI input fields D106 and D107 do not have validation in the ladder. A value of 0 produces 0 % Performance, and very large values could overflow the 32-bit arithmetic.
- **Integer arithmetic.** Divisions drop fractions, so displayed percentages can be up to 0.1 % lower than the exact value.
- **Scan order.** The OEE rungs share scratch registers D120 to D135, so they must stay in the program order shown and must not be changed without checking the register use.
- **Verification status.** A feature should only be described as working after it has been observed in the simulator.

13. Learning Outcomes

Through the work of Day 5 the following skills were practised:

- Planning a PLC device map and keeping it in a single device list.
- Writing ladder logic using SET/RST latches, rising-edge contacts and PLS pulses.
- Using SM402 for start-up initialisation and SM412 for a 1-second time base.
- Using 32-bit and 64-bit arithmetic (MUL, DMUL, DDIV) and per-mille scaling for accurate OEE calculation.
- Protecting a calculation against division by zero.
- Designing a four-screen HMI in GT Designer3 with lamps, switches, numerical displays, numerical inputs and an alarm viewer.
- Linking HMI objects to PLC devices and checking the project against its documentation.

14. Conclusion and Future Scope

14.1 Conclusion

An Industrial OEE Monitoring and Control System has been designed for a simulated conveyor line using a Mitsubishi FX5U PLC and a GOT HMI. The ladder program covers initialisation, Auto/Manual mode selection, start and stop control, fault detection with latch and buzzer, product and reject counting, time measurement, target detection and the calculation of Availability, Performance, Quality and OEE with 32-bit arithmetic and protection against division by zero. The HMI provides a Home screen, a Production Monitor screen, a Settings/Modes screen and an Alarm screen with an alarm viewer, all linked to the PLC devices in the device list.

The program and screens have been reviewed against the device list. A number of points listed in Section 11, especially the STOP button binding, should be verified in the simulator before the system is declared fully working. The full test plan in Section 10.3 should be completed and its results recorded.

14.2 Future scope

- Complete and record the full test plan, including OEE results against hand calculations.
- Replace the test bits M105, M106 and the fault bits M101, M107, M108 with the physical inputs X0 to X4 listed in the device list when hardware is available, and remove the temporary HMI test buttons.
- Delete the unused devices listed in Section 5.8 from the program.
- Add validation of the Ideal Cycle Time and Target inputs.
- Review the use of M107 as both a fault input and an over-target comparison coil.
- Add the reset confirmation popup screenshot and complete the alarm history configuration.

15. References

1. Mitsubishi Electric, MELSEC iQ-F FX5 User's Manuals (Application, Programming), for the FX5U instructions and special relays.
2. Mitsubishi Electric, GX Works3 Operating Manual.
3. Mitsubishi Electric, GT Designer3 (GOT2000) Screen Design Manual.
4. S. Nakajima, Introduction to TPM: Total Productive Maintenance, Productivity Press, 1988 (origin of the OEE concept).